Business Requirements Document (BRD)

Project Name: Coffee Logic

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1. Introduction

1.1 Purpose of the Document

This Business Requirements Document (BRD) is prepared to formally establish the objectives, strategic scope, and core requirements for the **Coffee Logic Ecosystem**. It serves as our primary roadmap, ensuring that the final digital product is perfectly aligned with our business vision of bringing world-class coffee directly to the doorsteps of our people.

This document is the "source of truth" for our Clients and our technical team. It ensures that as we move into the development phase, no part of the original vision is lost or diluted.

1.2 Scope of the Project

The scope of the **Coffee Logic Ecosystem** is centered on establishing a robust, end-to-end digital pipeline for the specialty coffee business. We are focusing our efforts on the core pillars of customer experience, automated recurring revenue, and precise inventory control.

I. In-Scope (What we shall deliver):

- A high-performance web interface allows customers to filter products by sensory attributes (Origin, Roast Level, Tasting Notes) and purchase either via one-time checkout or recurring subscription.
- A streamlined payment flow that recognizes returning customers and offers a "Default Payment" option to reduce friction and honor the customer's time.
- A background engine to handle recurring billing cycles and automated order generation without requiring manual intervention from the business owner.
- A secure Administrative Dashboard where the roastery can manage stock by specific "Roast Batches," ensuring the website always reflects the most recent roast dates.
- Automated email systems to notify customers of successful purchases, subscription renewals, and shipment updates.

II. Out-of-Scope (What is reserved for future phases):

- **Mobile Application:** This phase focuses exclusively on a "Mobile-Responsive" web platform. Native iOS or Android applications are excluded for the current rollout.
- **In-House Logistics:** The system will generate order data for fulfillment, but the actual physical delivery and fleet management are outside the software scope.
- **Third-Party Marketplace Integration:** For this phase, we are building our own "House of Coffee"; listing products on external sites like Amazon or Jumia is not included in this release.
- **Wholesale B2B Portal:** The current scope is strictly for D2C (Direct-to-Consumer). A separate portal for bulk hotel/cafe orders will be considered in Phase 2.

1.3 Document Overview

This document is not merely a list of technical tasks; it is a carefully woven narrative that guides us from the "Why" of our business to the "How" of our technology. We begin our journey in **Sections 2 and 3**, where we lay down the **Strategic Foundation**. Here, we speak of our ambitions—our "Business Goals"—and the markers of success that will prove we have built something of lasting value. We acknowledge our starting point as a fresh, digital-first entry into the market, ensuring we are all looking at the same horizon.

From there, we move into the "Heart of the Machine" in **Section 4**, which details our **Functional Requirements**. This is where we describe how the system breathes. We have organized this into five specialized modules: the *Sensory Catalog* that showcases the soul of our coffee; the *Dual-Action Commerce Engine* that manages our sales; the *Intelligent Checkout* that honors the customer's time; the *Admin Inventory Panel* that keeps our roasting batches in check; and the *Notification Suite* that serves as the automated voice of our brand.

In **Section 5**, we translate these requirements into **Use Cases**—real-world stories that show how our customers and roasters will actually walk through the system to achieve their goals. This is followed in **Section 6** by our **Foundational Standards of Excellence**. This essay outlines our "Non-Functional" promises: the speed of the interface, the absolute security of our people's data, and the premium "feel" of the mobile experience that ensures we are building for the long-term harvest.

Finally, in **Sections 7 through 9**, we deal with the **Realities of the Venture**. We define our "Constraints"—the boundaries of our budget and our respect for the timeline—while identifying the "Risks" and "Assumptions" we must manage to prevent any "wahala" during development. By the time you reach the end of this document, you will see a complete picture of a system that is secure, scalable, and ready to bring the finest coffee to the doorsteps of our people.

2. Business Objectives

2.1 Business Goals

The Coffee Logic project is anchored on four primary strategic pillars. These goals ensure that our technical efforts result in tangible commercial success and a premium brand reputation.

- Our foremost goal is to transition from a purely transactional model to a relationship-based one. By perfecting the subscription engine, we aim to secure a predictable and steady monthly cash flow, allowing the business to plan its roasting schedules with confidence.
- In today's fast-paced digital economy, we aim to honor our customers' time. By implementing intelligent checkout systems that remember user preferences and payment methods, we seek to eliminate all unnecessary hurdles in the purchasing journey, thereby increasing our conversion rates.
- We aim to position Coffee Logic as an authority in the market. By allowing customers to filter and discover coffee through sophisticated sensory profiles (Acidity, Body, and Notes), we are educating our market and building a community of loyal coffee connoisseurs rather than just casual buyers.
- We aim to automate the "donkey work" of the business. By automating stock updates and subscription renewals, the business owners can spend less time on paperwork and more time on the craft of roasting and expanding the reach of the brand.

2.2 Project Objectives

Our technical mandate is to engineer a sophisticated, dual-action commerce engine that allows our patrons to navigate between bespoke one-time purchases and recurring subscriptions with absolute ease. We shall anchor this experience in a world-class, tokenized checkout system, honoring the customer's time by enabling a secure "One-Click" return journey that is truly the hallmark of modern digital trade. Recognizing the realities of our environment, the platform shall be optimized for high performance even in low-bandwidth settings, ensuring that the premium Coffee Logic aesthetic remains accessible to all. Finally, we shall maintain an open and honest line of communication through an automated notification suite, keeping our people informed of their subscription status and the journey of their roast from our warehouse to their doorstep.

2.3 Success Criteria

The platform must achieve a 99.9% uptime during the first six months of operation, particularly ensuring that automated subscription billing occurs on the exact scheduled dates.The "Time-to-Checkout" for a returning customer should be reduced to under 15 seconds, proving that our "Intelligent Checkout" objective has been met.

3. Current Business Environment

As Coffee Logic is a fresh and ambitious entry into the specialty coffee market, there are no existing manual or legacy systems currently in operation. We are establishing this company as a **Digital-First E-Commerce Platform** from its very inception.

4 Functional Requirements

4.1. Sensory Catalog and Product Architecture

Governs how the Coffee is presented to the patron, ensuring the logic of coffee is clear.

ID	Requirement Name	Description (User Story Format)	Priority	Stakeholders	Objective	Success Metric
100	**Multi-Variant-display**	*The system shall allow users to select from various weight increments(200gm,500gm),Origin types(Brazilian,Ethiopian) and grind types(whole bean,Aeropress,Espresso)*	High	Customers, Roastery Ops, Inventory Management	Provide a highly customized shopping experience and ensure the correct product is roasted/ground.	< 5% error rate in fulfillment (grind/weight mismatches).
102	**Sensory Metadata-Engine**	*For every product there will be specification roast level acidity body specific tasting Notes*	High	Customers, Marketing, Quality Control	Increase customer confidence in purchasing premium beans by educating them on the flavor profile	Get positive feedback from atleast 40% of users .
103	**Dynamic-Pricing Logic**	*The system shall calculate and display two distinct price the standard price and discounted price*	High	Customers, Marketing, Finance	Use psychological pricing to incentivize subscriptions and clear stock of specific variants.	20% increase in new customer acquisition within the first month.

Module 2: Dynamic Dual Action Commerce Engine

This is the heart of "Coffee Logic" handling the switch between one-time sales and recurring revenue.

ID	Requirement Name	Description (User Story Format)	Priority	Stakeholders	Objective	Success Metric
200	Mode Selection Toggle	*The system shall provide a seamless UI toggle on the product detail page to switch between "one time purchase" and "regular subscription "*	High	Customers, Marketing, Frontend Dev.	Drive subscription "opt-in" rates by highlighting savings and convenience at the point of purchase.	At least 25% of all coffee sales converted to "Subscription" mode.
201	**Subscription Frequency Controller**	*If "subscription" is selected the system shall require to choose the delivery Candence (E.g every 7,14,or a month)*	High	Logistics/Shipping, Inventory Manager	Align delivery schedules with actual customer consumption habits to prevent "subscription fatigue."	< 10% of users requesting manual date changes via support.
203	**Subscription Lifecycle Management**	*The system **shall** allow authenticated users to pause, resume, or cancel their subscriptions through a secure "My Account" portal.*	High	Customers, Customer Support Team	Reduce support overhead and increase trust by giving the user full control over their commitment.	50% reduction in "Cancellation/Pause" related support tickets.

Module 3: The Intelligent Checkout (Payment & Security)

This module focuses on honoring the customer's time and securing their data.

ID	Requirement Name	Description (User Story Format)	Priority	Stakeholders	Objective	Success Metric
300	Tokenized Payment Vaulting	*The system **shall** integrate with Stripe to securely store payment tokens, ensuring the business never handles raw credit card data.*	High	Customers, DevOps, Legal/Compliance	Minimize PCI-DSS compliance risk and protect customer data from breaches.	0 instances of raw card data (PAN) stored in the local database.
301	**Default Payment Recognition**	*For returning users, the system **shall** automatically retrieve the saved "Default Card" and offer a "One-Click" checkout experience.*	High	Customers, Marketing Team	Reduce checkout friction and decrease "cart abandonment" rates.	Average checkout time for returning users under 15 seconds.
303	**Payment Pivot Logic**	*he checkout interface **shall** provide a clear option for users to "Add New Card" or "Change Default Payment" before finalizing an order..*	High	Customers, Customer Support	Provide flexibility and prevent failed transactions due to expired or maxed-out default cards.	< 5% support tickets related to "payment update" difficulties.
304	**Automated Billing Execution**	*For subscriptions, the system **shall** automatically trigger the payment at the defined interval without requiring a manual action from the customer.*	High	Finance Dept, Customers, Logistics	Ensure predictable recurring revenue and eliminate manual billing overhead.	99.9% of subscription renewals trigger automatically on the scheduled date.

Module 4: The Batch-Centric Inventory Panel (Admin)
This allows the business to manage the physical reality of the roastery.

ID	Requirement Name	Description (User Story Format)	Priority	Stakeholders	Objective	Success Metric
400	Batch Entry Management	*As an Admin, I want to log new batches of roasted coffee so that inventory levels are accurate*	High	Roastery Ops, Inventory Manager	Ensure real-time visibility of available stock to prevent overselling	100% accuracy between physical stock and digital inventory
401	**Depletion Safeguards**	*Once the total stock across all active batches hits zero, the system **shall** automatically disable the "Add to Cart" function and update the product status to "Roasting in Progress."*	High	Customers, Web Admin, Support	Prevent customer frustration and eliminate the need for manual stock "kill switches."	Zero "Out of Stock" orders placed by customers
403	**Order Fulfillment Dashboard**	*The Admin Panel **shall** provide a centralized view of all orders, categorized by status: Pending, Processing, Shipped, and Delivered.*	High	Fulfillment Team, Customer Support	Streamline the "order-to-ship" pipeline and reduce processing time.	Reduce average "Pending to Shipped" time by 20%.
404	**Revue Dashboard**	*To show the weekly, monthly and entire revenue and most sold type of coffee from the start of the ecosystem*	High	Business Owner, Marketing, Finance	Identify high-performing products and forecast future roasting needs.	Achievement of projected sales targets within the first 6 months

Module 5: The Automated Notification Suite

ID	Requirement Name	Description (User Story Format)	Priority	Stakeholders	Objective	Success Metric
500	Transactional Confirmations	*The system **shall** generate automated email/SMS receipts immediately upon successful payment.*	High	Customers, Finance Dept, Customer Support	Provide legal proof of purchase and reduce "did my order go through?" support tickets.	100% of successful payments trigger a notification within 30 seconds.
501	**Subscription Renewal Alerts**	*The system **shall** notify the customer 48 hours before an automated subscription charge is executed.*	High	Customers, Marketing, Product Owner	Build brand trust and reduce "accidental" charge complaints and bank chargebacks.	< 2% chargeback rate related to "unauthorized" renewals.
403	**Dunning Notifications**	*In the event of a payment failure, the system **shall** politely notify the customer and provide a secure link to update their payment details.*	High	Customers, Billing Team, Dev Ops	Recover "at-risk" revenue and prevent involuntary churn.	15% increase in successful payment recovery within 24 hours of failure.

5. Use-Case

UC-100: Customize and Purchase Coffee

- **Requirement Links:** 100, 102, 103
- **Actor:** Customer
- **The Story:** The customer explores the "Sensory Metadata" (tasting notes), selects their specific "Multi-Variant" (200g, Espresso grind, Ethiopian origin), and sees the "Dynamic Price" update based on their choices.
- **Critical Detail:** The system must ensure the specific combination selected is actually in stock before allowing the "Add to Cart" action.

UC-200: Set Up Recurring Subscription

- **Requirement Links:** 200, 201
- **Actor:** Customer
- **The Story:** The user flips the "Mode Selection Toggle" from One-Time to Subscription. They select their "Frequency" (e.g., every 14 days).
- **Critical Detail:** The system must calculate the discounted subscription price and clearly communicate when the *next* bag will ship.

UC-300: Secure Checkout and Card Vaulting

- **Requirement Links:** 300, 301, 303
- **Actor:** Customer / Stripe API
- **The Story:** The customer enters card details or selects a "Default Saved Card." The system "Vaults" the data through Stripe to ensure PCI compliance.
- **Critical Detail:** For returning users, this should be a "One-Click" experience to reduce cart abandonment.

UC-400: Manage Inventory and Batch Safeguards

- Requirement Links: 400, 401
- **Actor:** Admin / Roastery Manager
- **The Story:** The Admin logs a new batch of 50kg of Brazilian beans. As customers buy, the stock drops. When it hits zero, the "Depletion Safeguard" automatically switches the front-end status to "Out of Stock."
- **Critical Detail:** This must happen in real-time to prevent "overselling" (selling coffee you don't have).

UC-403: Process Orders and Monitor Revenue

- **Requirement Links:** 403, 404
- **Actor:** Fulfillment Team / Business Owner
- **The Story:** The fulfillment team uses the dashboard to move orders from "Processing" to "Shipped." The Business Owner views the "Revenue Dashboard" to see if the "Steady Cash Flow" goal is being met.
- **Critical Detail:** Status changes here should trigger the "Transactional Confirmation" (UC-500).

UC-405: Automated Subscription Maintenance (Dunning)

- **Requirement Links:** 304, 403 (Dunning), 501
- **Actor:** System (Automated) / Stripe
- **The Story:** 48 hours before a charge, the system sends an alert. On the due date, it triggers the payment. If the payment fails, it sends a "Dunning Notification" with a secure link to update the card.
- **Critical Detail:** This must be 100% automated to ensure "Passive Income" for the business.

Use Case ID	Use Case Name	Primary Actor	Priority
UC-100	Customize & Purchase Coffee	Customer	High
UC-200	Manage Subscription Setup	Customer	High
UC-300	Secure Payment Processing	Customer/Stripe	High
UC-400	Admin Inventory Control	Admin	High
UC-403	Order & Revenue Tracking	Admin/Owner	Medium
UC-405	Recurring Billing & Dunning	System	High

5.0. Non-Functional Requirements

To ensure that the **Coffee Logic** platform stands as a true mark of quality in the market, we are building it on a foundation that is as strong and reliable as the coffee we roast. These standards are not just technical "boxes to tick"; they are our word to the customer that we value their time, their safety, and their trust.

Performance: A System That Moves With Spirit

In our world, time is a gift, and we must not waste it. A customer coming to buy their morning coffee should not be met with a "hanging" screen. We are making sure this platform is "light" and very sharp. Even if the network is misbehaving or the data is slow, our core pages will load in **under 2 seconds**. We shall use clever ways to optimize our high-resolution coffee photos so they look beautiful but do not "heavy" the user's mobile phone. The experience must be smooth, like a well-poured cup.

Security: Protecting Our People's Peace

The trust of our patrons is a sacred thing. We cannot allow their private information to be exposed to "wahala." Because of this, security is our highest priority. The system will follow the **PCI-DSS standard** by using "Tokenization." What this means is simple: we will never store a customer's raw credit card numbers on our own side. Everything is encrypted and locked tight using the best industry standards, so our customers can sleep with their two eyes closed, knowing their money is safe.

Reliability: A Promise That Never Fails

Because we are handling people's daily bread—their coffee subscriptions—the system must be "awake" and ready **99.9% of the time**. There is no room for the system to "go down" when a payment is due. Our backend logic is being built to be independent and strong; it will trigger those recurring payments exactly when they are due, without us having to go in and push it manually. It is a "set-it-and-forget-it" promise.

Usability: The Respectful Interface

We believe a system should be "wise" enough to be simple. We are building an interface that is so intuitive that a first-time user—whether they are a young student or an elder—can complete their purchase without asking for help. We will make sure the **Subscription Toggle** and **Payment Options** are big, clear, and easy to tap on any mobile screen. We are honoring the customer's intelligence by making the path straight and clear.

Scalability: Building for the Harvest

We are not just building for today; we are building for the plenty that is coming. Whether we have 10 customers or 10,000, the system must not "tire." We have designed the database to be "future-proof" so that as we add more coffee origins and more batches, the speed of the **Inventory Panel** stays as fast as it was on day one. We are preparing the ground now for the great harvest ahead.

Maintainability: Good Stewardship of the Code

Lastly, we believe in doing a neat job. The code behind this platform will be clean and well-explained. This is our "stewardship"—ensuring that as the business grows, any professional developer can come in, understand the work, and add new features without "scattering" what is already there. We are building a digital asset that will serve this business for many years to come.

6. Constraints: Our Boundaries for Success

In every great venture, there are boundaries that keep the work on the right path. For **Coffee Logic**, we have identified three main areas where we must be disciplined to ensure the project finishes strong.

6.1 Budgetary Constraints: Strategic Spending

We are approaching this project with a "low-waste" mindset. To keep the foundation affordable, we will build upon **open-source software**, ensuring we aren't paying unnecessary license fees for things we can get for free. However, we must be realistic: a premium system requires some "fuel" to run. There will be modest costs for **cloud deployment** (to keep the site fast) and specific **API keys** (like Stripe for payments). We are committed to being good stewards of your funds, ensuring every cent spent brings a clear return to the business.

6.2 Timeline Constraints: Respecting the Season

We have set our faces toward a very clear goal. The first phase of this project—the "Heart" of the system—must be completed by the **end of February**. We are working with speed because we know the market does not wait. Following that, the full "harvest" of the project, including all advanced features and final polishing, will be delivered by the **end of March**. We are managing our hours carefully to ensure we meet these dates without compromising the quality of the work.

6.3 Regulatory and Compliance Constraints: Walking the Right Path

We are not just building for today; we are building to be "correct" in the eyes of the law. The system shall adhere to all **Digital Trade and Financial Regulations**. This means we are committed to **PCI-DSS compliance** for handling payments and following local **Data Protection Acts** (like NDPR or GDPR where applicable). We want to make sure the business is standing on a legal foundation that is as solid as a rock, protecting you from any future "wahala" or legal hiccups.